Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A model target needs a 3 km sideways shift over 5×10^{6} s. Its mass is 7×10^{8} kg. Estimate Δv and impulse; state the model used.

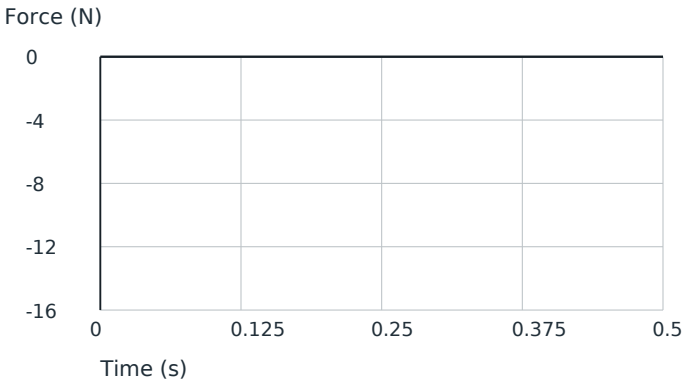
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$. Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

Worked reasoning: A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.



Marzano check - circle: ① ② ③

03 Read and interpret the model

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest. Draw before/after and explain which quantity is conserved.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart at +6 m/s sticks to a 1 kg cart at -3 m/s. Find final velocity and kinetic energy transformed. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart has momentum -24 kg·m/s and mass 4 kg. Find velocity and its new momentum if velocity reverses at unchanged speed.
Use before and after snapshots to organize the momentum account.

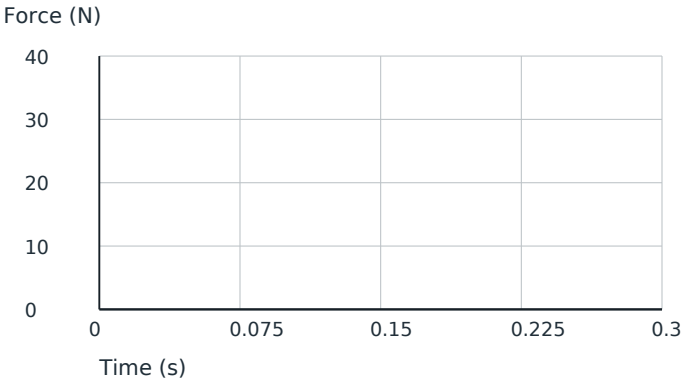
Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

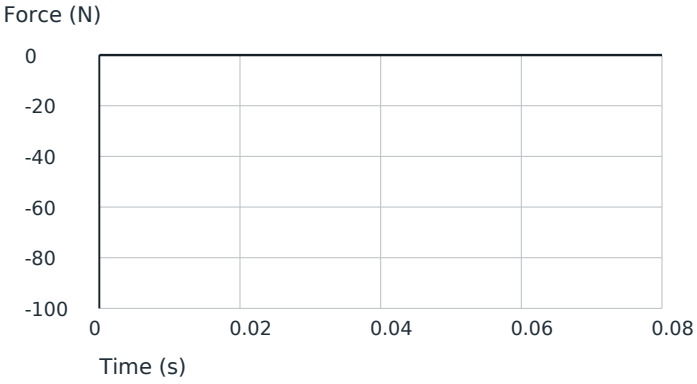
A triangular force pulse has peak 40 N and duration 0.30 s. It acts on a 2 kg cart initially at -1 m/s. Find impulse and final velocity. Organize the known quantities on the before-and-after model.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.4 kg ball rebounds from +5 to -3 m/s. A triangular force pulse lasts 0.08 s. Find impulse, average force and peak-force magnitude. Organize the known quantities on the before-and-after model.



Marzano check - circle: 1 2 3